TECHWISE

Smart E-Commerce & Product Intelligence Platform

PROJECT PLANNING & MANAGEMENT

Falcon Team
DEPI Graduation Project
Planning Window: 2–16 October 2026
Submission Deadline: 16 October 2026

WORKING DRAFT — REVIEW REQUIRED

Prepared from confirmed TechWise project context and the supplied DEPI guidelines. Proposed assignments and KPI targets require team approval.

1. Document Control

Field	Value
Project	TechWise — Smart E-Commerce & Product Intelligence Platform
Team	Falcon Team
Document	Project Planning & Management
Planning window	2 October 2026 – 16 October 2026
Submission deadline	16 October 2026
Status	Working draft; proposed assignments and KPI targets require team review

2. Executive Summary

TechWise is a smart e-commerce and product-intelligence platform designed to support the product lifecycle beyond a conventional online store: Discover → Understand → Compare → Decide → Buy → Own → Upgrade → Resell → Transfer.

The platform combines AI-assisted shopping, product intelligence, compatibility information, purchase decision support, product-truth mechanisms, lifecycle management, and a verified second-life marketplace.

Planning principle: AI supports intent understanding, explanation, recommendation, and summarization. Deterministic backend services and database records remain responsible for validation, business rules, compatibility decisions, calculations, authorization, ownership, and factual product state.

3. Project Proposal

3.1 Problem Statement

Conventional e-commerce experiences often focus on product discovery and checkout. Buyers may still need to research specifications, compare alternatives, understand compatibility, evaluate product claims, and decide whether to buy, upgrade, or resell. Second-life transactions also need clear eligibility and ownership-history rules. TechWise aims to bring these decision and lifecycle activities into one coherent platform.

3.2 Project Objectives

- Help users discover products through structured search, filtering, and AI-assisted natural-language intent understanding.
- Present product information to support understanding and comparison.
- Support purchase decisions using explicit requirements, compatibility information, and deterministic decision logic.
- Separate recorded product facts from AI-generated explanations and recommendations.
- Support a second-life marketplace subject to platform-origin, current-ownership, listing-eligibility, and dispute checks.
- Preserve product-asset history across resale while changing ownership records and protecting previous owners' personal information.

3.3 Scope — In Scope

- AI Shopping Intelligence.
- Verified Second-Life Marketplace.
- Product Compatibility Graph.
- Purchase Decision Engine.
- Product Truth Layer.
- Buy vs Upgrade vs Resell Engine.
- Advanced Smart Filtering as a cross-cutting capability, not a separate core feature.
- Responsive UI planning for desktop and mobile screen variants.

3.4 Scope — Not Yet Confirmed / Exclusions

- Specific payment gateway, hosting provider, AI model/provider, database engine, framework, or authentication provider unless formally selected.
- Claims of physical platform verification unless a corresponding process is implemented.
- Unapproved features beyond the confirmed core scope.
- Treating AI output as authoritative product facts, eligibility, ownership, or compatibility.

3.5 Primary Stakeholders

- End customers: discover, compare, decide, buy, own, upgrade, and resell products.
- Second-life sellers: list eligible products and provide required evidence.
- Second-life buyers: evaluate condition, provenance, and available evidence.
- Platform administrators/moderators: review listings, disputes, and policy compliance.
- Falcon Team: plan, design, implement, test, document, and present the system.
- DEPI instructors/evaluators: assess planning, requirements, design, implementation, testing, documentation, and presentation.

4. Project Plan

4.1 Delivery Approach

The team will work cross-functionally. Work is divided by screens, features, modules, and deliverables rather than isolating members into permanent UI-only, backend-only, database-only, or testing-only silos. Each task should have one accountable coordinator/reviewer, while implementation and validation may involve multiple members.

4.2 Schedule and Milestones

Workstream	Start	Deadline	Milestone / evidence
Project Planning & Management	2 Oct 2026	16 Oct 2026	Proposal, plan/Gantt, roles, risk plan, KPIs uploaded to GitHub
Literature Review	9 Oct 2026	16 Oct 2026	Relevant prior work and evaluation/improvement notes documented
Requirements Gathering	9 Oct 2026	16 Oct 2026	Stakeholders, user stories/use cases, functional and non-functional requirements
System Analysis & Design	9 Oct 2026	6 Nov 2026	Architecture, required diagrams, data model, UI/UX prototypes and guidelines
Implementation	16 Oct 2026	30 Nov 2026	Integrated source code and executable/deployed demonstration as feasible
Final Presentation, Testing & Reports	16 Oct 2026	4 Dec 2026	Test evidence, user/technical docs, presentation and final report

4.3 Near-Term Work Plan

Date	Activity	Exit condition
10 Oct	Inventory existing documents and decisions	Existing work classified complete / partial / missing
10–12 Oct	Complete proposal, scope, schedule, coordination model	Draft reviewed by team lead and members
12–14 Oct	Complete risk register and measurable KPIs	Risk owners and measurement methods defined
14–15 Oct	Consistency review and GitHub packaging	Dates, scope, terminology consistent; files readable
16 Oct	Submit planning package and other due phase deliverables	Required documents uploaded; repository link checked

4.4 Resource Allocation

Primary resources are the six Falcon Team members, their development/design tools, the shared GitHub repository, project documentation, and UI prototypes in progress. Specific technology choices and external service costs remain unconfirmed and must not be presented as committed resources until approved.

5. Task Assignment & Roles

Operating model: all six members contribute across UI/UX, frontend, backend, data, testing, integration, and documentation according to task needs. Assignments below define coordination/accountability, not exclusive technical silos. Proposed coordination assignments must be reviewed and accepted by the named members before being treated as final.

Member	Coordination responsibility (proposed)	Shared contributions
Elshafei Mohamed (Team Leader)	Overall planning, scope and decision tracking; AI module ownership; integration follow-up; final consistency review	Cross-functional implementation, testing, documentation, presentation
Hagar Atef	Coordinate an assigned feature/screen package and track acceptance criteria	UI/UX, implementation, testing, documentation, integration
Shahd Mohamed	Coordinate an assigned feature/screen package and track acceptance criteria	UI/UX, implementation, testing, documentation, integration
Samer Walid	Coordinate an assigned feature/screen package and track acceptance criteria	UI/UX, implementation, testing, documentation, integration
Mostafa Mohamed	Coordinate an assigned feature/screen package and track acceptance criteria	UI/UX, implementation, testing, documentation, integration
Mohamed Alaa El-Din Fathallah Nafea	Coordinate an assigned feature/screen package and track acceptance criteria	UI/UX, implementation, testing, documentation, integration

Important: specific feature/screen ownership for the five non-lead members is intentionally left for team confirmation rather than guessed. Once the UI/screen inventory and module boundaries are finalized, assign a task-level owner, reviewer, due date, and acceptance criteria to each item.

5.1 Team Working Rules

- Use one branch per person plus main, as previously chosen; do not introduce a develop branch unless the team explicitly changes this decision.
- Break work into reviewable tasks with an owner, deadline, dependencies, and acceptance criteria.
- Keep main in a reviewable state and use the team's agreed review process.
- Record material scope, architecture, data, AI-boundary, and business-rule decisions in project documentation.
- Do not mark a task complete until its output has been reviewed and relevant evidence is available.

6. Risk Assessment & Mitigation Plan

ID	Risk	Likelihood	Impact	Mitigation / response	Owner
R1	Overlapping planning and requirements deadlines reduce review time	High	High	Prioritize deadline-bound documents; short check-ins; set review cutoff before submission	Team Leader + all members
R2	Scope grows beyond team capacity	Medium	High	Protect six confirmed core features; treat additions as proposals requiring approval	Team Leader
R3	UI prototypes are mistaken for implemented/integrated features	Medium	High	Track design, implementation, integration, and testing as separate statuses	Feature coordinators
R4	AI-generated statements are treated as factual product data	Medium	High	Require structured outputs and backend validation; facts come from product records or verified sources	AI Module Owner + backend contributors
R5	Compatibility or decision logic differs across UI and backend	Medium	High	Keep validation, ranking, compatibility, and calculations in	Feature coordinators

				deterministic services; define contracts and tests	
R6	Second-life listing bypasses origin/ownership/eligibility checks	Medium	High	Validate platform origin, current ownership, listing state, eligibility, and blocking disputes	Second-Life coordinator (to assign)
R7	Previous owner's personal data is exposed after resale	Low–Medium	High	Separate ownership history from personal data; do not expose previous owners' private information	Team Leader + data/security contributors
R8	Late integration causes conflicting interfaces or broken flows	Medium	High	Agree contracts early; integrate continuously; maintain blocker list and regression checks	All members
R9	Unconfirmed technologies are documented as final decisions	Medium	Medium	Label undecided choices pending; approve stack/service decisions before marking final	Team Leader + team
R10	GitHub documentation is stale or missing	Medium	Medium	Keep one source of truth; check file links and dates before each milestone	Documentation coordinator (to assign)

7. Key Performance Indicators (KPIs)

The following are proposed project-management and product-quality KPIs. Numerical targets are draft targets for team approval, not DEPI-mandated values.

KPI	Proposed target	Measurement	Review point
Planning deliverable completion	100% of planning artifacts reviewed by 15 Oct	Checklist: proposal, plan, roles, risks, KPIs	Before submission
Milestone adherence	At least 90% of planned tasks completed by due date	Tasks completed on time ÷ tasks due	Weekly
Task traceability	100% of active tasks have owner, due date, acceptance criteria	Review task list/repository issues	Weekly
Integration health	No known release-blocking issue unresolved at milestone review	Blocker list and integration test results	Milestone reviews
Core-flow test coverage	All agreed critical journeys have documented test cases before final testing closes	Test plan mapped to core features	Implementation/testing
AI factual-boundary compliance	All AI-dependent flows define validation/source-of-truth rules before acceptance	Review AI contracts and tests	Design/integration
Second-life eligibility	All listing acceptance tests cover origin, current ownership, listing state, eligibility, disputes	Requirements-to-test traceability	Feature acceptance
Responsive design coverage	Every approved screen has desktop/mobile counterpart except explicitly mobile-only screens	Compare screen inventory with prototypes	Design review
Documentation freshness	All milestone documents updated before related submission	Repository and commit review	Every milestone

8. Monitoring, Communication & Change Control

8.1 Progress Monitoring

- Maintain a shared task list with statuses: Not Started, In Progress, Blocked, In Review, Done.
- Review progress, blockers, dependencies, and deadlines at least twice per week during the planning window.

- Use evidence-based status: a design is not implementation; local implementation is not integration; a passing test applies only to tested scope.

8.2 Change Control

- Record proposed changes to scope, deadlines, responsibilities, data rules, AI boundaries, and architecture.
- Assess impact on schedule, dependencies, testing, and documentation before accepting a change.
- Only explicit team decisions convert a proposal into a confirmed decision.
- Update affected documents and the decision log after approval.

8.3 Communication and Repository

- GitHub is the shared destination for project documents, as required by the supplied DEPI guidance.
- Use meaningful commit messages and keep document versions identifiable.
- At each submission, verify repository URL, file paths, readability, and latest commit.
- Repository URL: insert the actual URL; it was not available in the context used for this draft.

9. Acceptance Checklist

- Project Proposal: overview, objectives, and scope are explicit.
- Project Plan: timeline/Gantt representation, milestones, deliverables, and resource allocation are included.
- Task Assignment & Roles: all six members are named; proposed coordination assignments are confirmed.
- Risk Assessment: risks include likelihood, impact, mitigation, and owner.
- KPIs: each KPI has a definition, target, measurement method, and review point.
- Cross-document consistency: dates, feature names, ownership rules, AI boundaries, and scope agree.
- Repository: required files are uploaded to GitHub and the link is verified.
- Team review: placeholders and proposed targets are approved or explicitly marked pending.

10. Open Items Requiring Team Confirmation

- Confirm the exact DEPI track name and any instructor template/rubric.
- Insert the actual GitHub repository URL and folder structure.
- Approve task-coordination assignments for the five non-lead members.
- Approve or revise proposed KPI targets.
- Confirm formally approved technology-stack decisions; do not infer them from prototypes.
- Confirm whether the instructor expects one combined document or separate files for each planning artifact.